

CONSTRAINED CHOICE: THEORY AND APPLICATIONS

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- [GHS24]** Galichon, A., Y.-W. Hsieh, and M. Sylvestre (2024): “Monotone comparative statics for submodular functions, with an application to aggregated deferred acceptance,” *Revise and resubmit*, Journal of Economic Theory. [arXiv:2304.12171](#).
- [GHJ]** Galichon, A., Y.-W. Hsieh, and A. Jacquet: “Aggregate Stable Matching with Money Burning,” *Journal of Mechanism and Institution Design*, Special Issue in Memory of YingHua He.
- [G26]** Galichon, A. (2026): *Discrete Choice Models: Mathematical Methods, Econometrics, and Data Science*, Princeton University Press; in particular, ch. 7 and app. A.3.6.

- ▶ In many real-world choice problems, agents do not face an unconstrained menu: hospitals have capacity, schools have seats, taxi drivers can be busy, contracts can be unavailable.
- ▶ Whenever demand exceeds supply, a *rationing* device is needed. Economists are familiar with one such device — prices — but in many markets prices are absent or rigid:
 - ▶ school choice, college admissions, organ allocation;
 - ▶ public housing, refugee resettlement;
 - ▶ labor markets with rigid wages.
- ▶ In their absence, congestion appears in the form of *shadow prices*: waiting times, lower probability of admission, money burned.
- ▶ This talk: a unified theory of **constrained choice**, where the rationing variable τ may or may not be a Lagrange multiplier, and where existence/uniqueness/comparative statics flow from a single mathematical engine: the theory of **M-functions**.
- ▶ Applications: NTU matching with random utility, aggregate deferred acceptance, money burning, divisible-good extensions.

1. Constrained choice as welfare maximization [G26, §7.2]
2. Constrained choice with general choice functions [G26, §7.1]
3. M-functions: existence and uniqueness [G26, App. A.3.6]
4. The logit illustration [G26, §7.1]
5. Comparative statics [G26, §7.3]
6. NTU matching with random utility [G26, §7.5]
7. Deferred acceptance [G26, §7.6]
8. Reinterpretation à la Hatfield–Milgrom [G26, §7.7]
9. The set-valued case [GHS24]

Section 1

CONSTRAINED CHOICE AS WELFARE MAXIMIZATION

We start from a discrete choice setting: Y alternatives plus an outside option, systematic utilities $U \in \mathbb{R}^Y$, and an unconstrained welfare function

$$G(U) = \mathbb{E} \left[\max_{y \in [Y]} \{U_y + \varepsilon_y, \varepsilon_0\} \right], \quad G^*(\pi) = \max_U \{U^\top \pi - G(U)\}.$$

Now suppose alternative y has capacity $\bar{\pi}_y$. The *constrained welfare function* is

$$\tilde{G}(U, \bar{\pi}) = \max_{\pi \in \mathbb{R}_+^Y} \left\{ \sum_{y \in [Y]} \pi_y U_y - G^*(\pi) : \pi \leq \bar{\pi} \right\}.$$

The constrained market share map $\tilde{\pi}(U, \bar{\pi})$ is the solution; the dual variables associated with $\pi \leq \bar{\pi}$ are, *by definition*, Lagrange multipliers.

The Lagrangian

$$\mathcal{L}(\tau, \pi) = \sum_y \bar{\pi}_y \tau_y + \sum_y \pi_y (U_y - \tau_y) - G^*(\pi)$$

admits the dual representation

$$\tilde{G}(U, \bar{\pi}) = \min_{\tau \geq 0} \left\{ \sum_y \bar{\pi}_y \tau_y + G(U - \tau) \right\}.$$

Under continuity and full-support assumptions on ε :

$$\tilde{\pi}(U, \bar{\pi}) = \nabla_U \tilde{G}(U, \bar{\pi}), \quad \tau(U, \bar{\pi}) = \nabla_{\bar{\pi}} \tilde{G}(U, \bar{\pi}).$$

Reading. τ_y is the marginal welfare gain from relaxing the capacity constraint on alternative y ; the constraint is binding iff $\tau_y > 0$.

With i.i.d. draws $\varepsilon_{iy} \sim \mathcal{P}$, $i \in [I]$, $y \in [Y]$, the constrained welfare becomes a linear program:

$$\begin{aligned} \tilde{G}_I(U, \bar{\pi}) = & \max_{\lambda \in \mathbb{R}_+^{I \times (Y+1)}} \sum_{iy} \lambda_{iy} (U_y + \varepsilon_{iy}) \\ \text{s.t.} \quad & \sum_{y \in [Y] \cup \{0\}} \lambda_{iy} = 1/I, \quad [u_i] \\ & \sum_{i \in [I]} \lambda_{iy} \leq \bar{\pi}_y, \quad [\tau_y \geq 0] \end{aligned}$$

with dual

$$\begin{aligned} \tilde{G}_I(U, \bar{\pi}) = & \min_{u \in \mathbb{R}^I, \tau \in \mathbb{R}_+^Y} \frac{1}{I} \sum_i u_i + \sum_y \bar{\pi}_y \tau_y \\ \text{s.t.} \quad & u_i \geq U_y - \tau_y + \varepsilon_{iy}, \quad [\lambda_{iy} \geq 0]. \end{aligned}$$

Both τ (capacity duals) and u (assignment duals) come out of any standard LP solver.

FROM [G26, APP. CODE]

```
import numpy as np, gurobipy as grb
np.random.seed(7)
I, Y = 100, 3
U_y = np.array([0.2, -0.3, 0.5])
pibar_y = np.array([0.3, 0.2, 0.1])
epsilon_i_y = np.random.normal(size=(I, Y+1))

m = grb.Model()
lambda_i_y = m.addMVar((I, Y+1))
m.setObjective(
    (lambda_i_y * (np.array([0]+list(U_y)))[None,:]) + epsilon_i_y).sum(),
    sense=grb.GRB.MAXIMIZE)
u_i = m.addConstr(lambda_i_y.sum(axis=1) == 1/I)
tau_y = m.addConstr(lambda_i_y.sum(axis=0)[1:] <= pibar_y)
m.optimize()

print('tau_y =', tau_y.pi) # Lagrange multipliers = shadow prices
```

Section 2

CONSTRAINED CHOICE WITH GENERAL CHOICE FUNCTIONS

- ▶ In the previous section, $\pi(U) = \nabla G(U)$ came from a welfare problem, and τ was a genuine Lagrange multiplier.
- ▶ More generally, we may only have a primitive *demand function* $\pi : \mathbb{R}^Y \rightarrow \mathbb{R}^Y$, not necessarily the gradient of a convex potential:
 - ▶ bounded-rationality models;
 - ▶ models with reference points or behavioral biases;
 - ▶ choice rules elicited from data without a structural derivation.
- ▶ We still want a meaningful notion of equilibrium under capacity:

$$\bar{\pi}_y - \pi_y(U - \tau) \geq 0, \quad \tau_y \geq 0, \quad \tau_y(\bar{\pi}_y - \pi_y(U - \tau)) = 0.$$

- ▶ Then τ is a *shadow price*, but no longer the Lagrange multiplier of an underlying maximization.

Setting: agent picks $y \in [Y]$ or opts out, utility $\max_y \{U_y + \varepsilon_y, \varepsilon_0\}$.

Demand $\pi_y(U)$ may exceed capacity $\bar{\pi}_y$. We model congestion by a utility shading $U_y \rightarrow U_y - \tau_y$.

SYSTEM FOR τ

$\tau \in \mathbb{R}^Y$ solves

$$\begin{cases} \tau_y \geq 0, & \forall y \in [Y] \\ \bar{\pi}_y - \pi_y(U - \tau) \geq 0, & \forall y \in [Y] \\ \tau_y(\bar{\pi}_y - \pi_y(U - \tau)) = 0, & \forall y \in [Y]. \end{cases}$$

THEOREM (EXISTENCE/UNIQUENESS)

Under continuity and full-support assumptions on the distribution of ε , the system above has a unique solution τ .

For each y , define the excess capacity

$$\rho_y := \bar{\pi}_y - \pi_y(U - \tau).$$

τ_y and ρ_y are nonneg. and complementary. Encode them in a single signed vector

$$\theta = \tau - \rho, \quad \tau = \theta^+, \quad \rho = \theta^-.$$

The system is then equivalent to the single (unconstrained) equation

$$\pi(U - \theta^+) + \theta^- = \bar{\pi}.$$

PROPOSITION

τ solves the original system iff $\tau = \theta^+$ where θ solves the boxed equation.

Define

$$\mathbf{q}(\theta) := \bar{\pi} - \theta^- - \pi(U - \theta^+).$$

Then θ solves our system iff $\mathbf{q}(\theta) = 0$.

PROPOSITION

Under continuity and full-support, the map $\mathbf{q}$ is an *M-function*.

PROPOSITION (EXISTENCE & UNIQUENESS)

Hence the solution θ exists and is unique, and so do $\tau(U, \bar{\pi}) = \theta^+$, the constrained market share $\tilde{\pi}(U, \bar{\pi}) = \pi(U - \theta^+)$, and the rejection map $\rho(U, \bar{\pi}) = \theta^-$.

We now turn to the engine: **the theory of M-functions**.

Section 3

M-FUNCTIONS: THE EXISTENCE/UNIQUENESS ENGINE

DEFINITION

$\mathbf{q} : \mathbb{R}^X \rightarrow \mathbb{R}^X$ is a *Z-function* if each $\mathbf{q}_x$ is nonincreasing in p_y for every $y \neq x$.

Synonymous: “off-diagonally antitone,” or “gross substitutes.”

Examples.

(I) $\mathbf{q}(p) = Ap$ with $A_{xy} \leq 0$ for $x \neq y$: A is a *Z-matrix*.

(II) For costs c_{xy} ,

$$\mathbf{q}_z(p) = \sum_x \exp(p_z - p_x - c_{xz}) - \sum_y \exp(p_y - p_z - c_{zy}).$$

(III) If $\mathbf{q}$ is a Z-function, so is $\tilde{\mathbf{q}}(p) := \mathbf{q}(p^+) - p^-$.

(IV) If $f : \mathbb{R}^X \rightarrow \mathbb{R}$ is differentiable and submodular, ∇f is a Z-function.

PROPOSITION

Let $\mathbf{q} : \mathbb{R}^X \rightarrow \mathbb{R}^X$ be differentiable. Then $\mathbf{q}$ is a Z-function iff its Jacobian $D\mathbf{q}(p)$ is a Z-matrix at every $p \in \mathbb{R}^X$.

DEFINITION (WEAK NONREVERSINGNESS)

$\mathbf{q}$ is *weakly nonreversing* if

$$p \leq p' \text{ and } \mathbf{q}(p) \geq \mathbf{q}(p') \implies \mathbf{q}(p) = \mathbf{q}(p').$$

DEFINITION (STRONG NONREVERSINGNESS)

$\mathbf{q}$ is *strongly nonreversing* if

$$p \leq p' \text{ and } \mathbf{q}(p) \geq \mathbf{q}(p') \implies p = p'.$$

Examples (selection). Isotone functions are weakly nonreversing. The gradient of a convex function is weakly nonreversing; if it is the gradient of a strictly convex function, strongly so. Inverse-isotone functions are strongly nonreversing.

DEFINITION

$\mathbf{q}$ is an **M0-function** if it is a Z-function and weakly nonreversing.

$\mathbf{q}$ is an **M-function** if it is a Z-function and strongly nonreversing.

THEOREM

Let u be a C^1 convex submodular function. Then ∇u is an M0-function. If u is in addition strictly convex, ∇u is an M-function.

This is exactly the structural result we need to apply to

$$\mathbf{q}(\theta) = \bar{\pi} - \theta^- - \pi(U - \theta^+).$$

THEOREM (M0)

Let $\mathbf{q}$ be a Z-function. TFAE:

- (I) $\mathbf{q}$ is an M0-function;
- (II) $\mathbf{q}^{-1}$ is increasing in the strong set order: if $\mathbf{q}(p) \leq \mathbf{q}(p')$ then $\mathbf{q}(p) = \mathbf{q}(p \wedge p')$ and $\mathbf{q}(p') = \mathbf{q}(p \vee p')$.

THEOREM (M)

Let $\mathbf{q}$ be a Z-function. TFAE:

- (I) $\mathbf{q}$ is an M-function;
- (II) $\mathbf{q}^{-1}$ is isotone on its domain: if $\mathbf{q}(p) \leq \mathbf{q}(p')$ then $p \leq p'$.

SUBSTITUTES

Weak substitutes: $\mathbf{q}$ is a Z-function ($\mathbf{q}_x$ nonincreasing in p_y , $y \neq x$).

Strong substitutes: $\mathbf{q}_x$ is *strictly* decreasing in p_y for $y \neq x$.

LAW OF AGGREGATE SUPPLY

Weak: $\sum_x \mathbf{q}_x$ is nondecreasing in each p_x .

Strong: $\sum_x \mathbf{q}_x$ is strictly increasing in each p_x .

Connected strong substitutes (intermediate). For any $B \subseteq [X]$ and any p, p' with $p_x = p'_x$ on B and $p_x > p'_x$ off B , there exists $x \in B \cup \{0\}$ with $\mathbf{q}_x(p) < \mathbf{q}_x(p')$.

THEOREM (BGH)

- (I) Weak substitutes + weak law of aggregate supply $\Rightarrow$ M0-function.
- (II) + connected strong substitutes $\Rightarrow$ M-function.
- (III) Weak substitutes + strong law of aggregate supply $\Rightarrow$ M-function.

Section 4

THE LOGIT ILLUSTRATION

Normalize $U_0 = 0$ and $\bar{\pi}_0 = 1$. Unconstrained logit: $\pi_y(U) = \frac{e^{U_y - u}}{e^{-u} + \sum_y e^{U_y - u}} = 1$.

Constrained version: $\tau_y > 0 \Rightarrow e^{U_y - \tau_y - u} = \bar{\pi}_y$, i.e. $\tau_y = U_y - u - \log \bar{\pi}_y$.
If $\tau_y = 0$, then $e^{U_y - u} \leq \bar{\pi}_y$, i.e. $U_y - u - \log \bar{\pi}_y \leq 0$. Hence:

CLOSED-FORM CONSTRAINED MARKET SHARES

$$\tilde{\pi}_y(U, \bar{\pi}) = \min(e^{U_y - u}, \bar{\pi}_y),$$

with $u \in \mathbb{R}$ determined by the scalar fixed-point equation

$$e^{-u} + \sum_{y \in [Y]} \min(e^{U_y - u}, \bar{\pi}_y) = 1.$$

A one-dimensional root-finding problem in u .

FROM [G26, APP. CODE]

```
import scipy.optimize as opt, numpy as np

U_y = np.array([0.2, -0.3, 0.5])
pibar_y = np.array([0.3, 0.5, 0.6])

# scalar root-finding for  $e^{-u}$ 
mu0 = opt.brentq(
    lambda m: m + np.minimum(m*np.exp(U_y), pibar_y).sum() - 1,
    0, 1)

u = -np.log(mu0)
print('u =', u)
print('pi_tilde =', np.minimum(mu0*np.exp(U_y), pibar_y))
```

A 3-line solver for the constrained logit, ready for use as a building block in DA below.

Section 5

COMPARATIVE STATICS

Intuition: tighter capacity $\bar{\pi}$ should yield higher shadow prices and more rejections.

Recall $\theta = \tau - \rho$ solves $\pi(U - \theta^+) + \theta^- = \bar{\pi}$.

If $\bar{\pi} \leq \bar{\pi}'$, the M-function property of $\theta \mapsto \bar{\pi} - \pi(U - \theta^+) - \theta^-$ implies $\theta \geq \theta'$.

THEOREM (ISOTONE RESPONSE)

Under continuity/full-support:

- (I) $\bar{\pi} \mapsto \tau(U, \bar{\pi})$ is **antitone**;
- (II) $\bar{\pi} \mapsto \rho(U, \bar{\pi})$ is **isotone**.

The theorem is a manifestation of **substitutability**: in discrete choice, agents pick exactly one option, so options must be substitutes.

- ▶ *Tighter constraints $\Rightarrow$ higher shadow prices.* Reduced supply on one option shifts demand to others, pushing them toward their capacity limits and lifting shadow prices everywhere.
- ▶ *Tighter constraints $\Rightarrow$ less unused capacity.* An offer rejected when constraints are loose remains rejected when constraints tighten: chosen offers continue to dominate.

These are the natural “capacity-induced response” results. They will reappear in much greater generality in the set-valued case.

Section 6

NTU MATCHING WITH RANDOM UTILITY

A matching market with *fixed wages*.

A firm i of type $x \in [X]$ matched with a worker of type y enjoys utility

$$a_{xy} + \varepsilon_{iy};$$

unmatched, ε_{i0} . The vector $(\varepsilon_{iy})_y \sim \mathbf{P}_x$ already *includes* the wage paid.

A worker j of type $y \in [Y]$ matched with a firm of type x enjoys

$$\gamma_{xy} + \eta_{jx};$$

unmatched, η_{j0} , with $(\eta_{jx})_x \sim \mathbf{Q}_y$.

Goal. Define an equilibrium notion when prices cannot adjust to clear the market.

Systematic utilities $U, V \in \mathbb{R}^{X \times Y}$. The unconstrained welfare functions are

$$G(U) = \sum_x n_x \mathbb{E}_{\mathcal{P}_x} \left[\max_y \{ U_{xy} + \varepsilon_y, \varepsilon_0 \} \right],$$

$$H(V) = \sum_y m_y \mathbb{E}_{\mathcal{Q}_y} \left[\max_x \{ V_{xy} + \eta_x, \eta_0 \} \right],$$

with conjugates (entropies of choice)

$$G^*(\mu) = \max_U \left\{ \sum_{xy} \mu_{xy} U_{xy} - G(U) \right\},$$

$$H^*(\mu) = \max_V \left\{ \sum_{xy} \mu_{xy} V_{xy} - H(V) \right\}.$$

And the constrained versions, under capacity $\mu \leq \bar{\mu}$:

$$\tilde{G}(U, \bar{\mu}) = \max_{\mu \in \mathbb{R}_+^{X \times Y}} \left\{ \sum \mu_{xy} U_{xy} - G^*(\mu) : \mu \leq \bar{\mu} \right\}.$$

And similarly $\tilde{H}(V, \bar{\mu})$. Define $\tilde{\mu}^\alpha(\bar{\mu}) = \partial_U \tilde{G}(\alpha, \bar{\mu})$, $\tau^\alpha(\bar{\mu}) = \partial_{\bar{\mu}} \tilde{G}(\alpha, \bar{\mu})$, and analogously $\tilde{\mu}^\gamma$, τ^γ .

Without a market-clearing price, some types are overdemanded $\Rightarrow$ shadow prices $\tau_{xy}^\alpha, \tau_{xy}^\gamma \geq 0$ shading utilities.

If $\tau_{xy}^\alpha > 0$: firms of type x compete for workers of type y (x is the **long side** on xy); then $\tau_{xy}^\gamma = 0$. So $\min\{\tau_{xy}^\alpha, \tau_{xy}^\gamma\} = 0$ *everywhere*.

DEFINITION

$\mu \in \mathbb{R}_+^{X \times Y}$ is an *equilibrium matching without prices* if there exist $\tau^\alpha, \tau^\gamma \in \mathbb{R}_+^{X \times Y}$ with

$$\min\{\tau_{xy}^\alpha, \tau_{xy}^\gamma\} = 0 \quad \text{and} \quad \mu = \nabla G(\alpha - \tau^\alpha) = \nabla H(\gamma - \tau^\gamma).$$

THEOREM

Under continuity/full-support, an equilibrium matching without prices exists and is unique.

With i.i.d. Gumbel shocks on both sides, choice ratios give

$$\frac{\mu_{xy}}{\mu_{x0}} = e^{\alpha_{xy} - \tau_{xy}^{\alpha}}, \quad \frac{\mu_{xy}}{\mu_{0y}} = e^{\gamma_{xy} - \tau_{xy}^{\gamma}},$$

and $\min\{\tau^{\alpha}, \tau^{\gamma}\} = 0$ yields

$$\mu_{xy} = \min(\mu_{x0} e^{\alpha_{xy}}, \mu_{0y} e^{\gamma_{xy}})$$

with μ_{x0}, μ_{0y} pinned down by population margins.

Contrast with Choo–Siow (TU, transferable utility):

$$\mu_{xy} = \sqrt{\mu_{x0} \mu_{0y} e^{\alpha_{xy} + \gamma_{xy}}}.$$

TU: *geometric mean*, transfers absorb mismatch.

NTU: *minimum*, the short side rations.

Section 7

AGGREGATE DEFERRED ACCEPTANCE

At time $t = 0$, available offers $\mu_{xy}^{A,0} = \min(n_x, m_y)$ for every xy .

Period 1, proposal. If unconstrained, firms would propose $\mu^{P,1} = \nabla G(\alpha)$.

Period 1, disposal. If overdemand on the worker side, workers select using the constrained-choice map for type- γ welfare:

$$\mu^{K,1} = \tilde{\mu}^\gamma(\mu^{P,1}).$$

Period 1, update. Rejected offers are removed:

$$\mu^{A,1} = \mu^{A,0} - (\mu^{P,1} - \mu^{K,1}).$$

Period t , proposal. Subject to availability $\mu^{A,t-1}$:

$$\mu^{P,t} = \tilde{\mu}^{\alpha}(\mu^{A,t-1}).$$

Period t , disposal.

$$\mu^{K,t} = \tilde{\mu}^{\gamma}(\mu^{P,t}).$$

Period t , update.

$$\mu^{A,t} = \mu^{A,t-1} - (\mu^{P,t} - \mu^{K,t}).$$

The constrained-choice machinery makes *each step* a well-defined optimization that can be solved with the M-function tools or directly with an LP solver — exactly the modules built in Sections 1–4.

THEOREM (DA CONVERGENCE)

Under continuity/full-support, the sequences $\mu^{P,t}$ and $\mu^{K,t}$ both converge to an equilibrium matching without prices.

Proof (sketch, via 5 claims). Set $\tau_{xy}^{\alpha,t} = \tau_{xy}^{\alpha}(\mu^{A,t})$, $\tau_{xy}^{\gamma,t} = \tau_{xy}^{\gamma}(\mu^{P,t})$.

1. $\mu^{P,t+1} \geq \mu^{K,t}$: there cannot be fewer offers proposed than were just kept.
2. $\tau^{\alpha,t}$ is nondecreasing in t ; $\tau^{\gamma,t}$ is nonincreasing.
3. $\min\{\tau_{xy}^{\alpha,t}, \tau_{xy}^{\gamma,t}\} = 0$ for every t, xy .
4. Each of $\mu^{A,t}, \mu^{P,t}, \mu^{K,t}, \tau^{\alpha,t}, \tau^{\gamma,t}$ converges.
5. The limits constitute an equilibrium matching without prices. □

FROM [G26, APP. CODE]

```
proposers = LogitChoiceProblem(n_x, alpha_x_y)
responders = LogitChoiceProblem(m_y, gamma_x_y.T)

muA_x_y = np.minimum(n_x[:, None], m_y[None, :])
maxiter, rel_tol = int(1e4), 1e-3

for i in range(maxiter):
    muP_x_y = proposers.choice(constraint=muA_x_y)
    muK_x_y = responders.choice(constraint=muP_x_y.T).T
    muR_x_y = muP_x_y - muK_x_y
    if muR_x_y.sum() < rel_tol:
        break
    muA_x_y = muA_x_y - muR_x_y

print('Converged in', i, 'steps; matched =', muK_x_y.sum())
```

The same loop runs over individual, logit, or simulated heterogeneity — only the ChoiceProblem subclass changes.

Section 8

REINTERPRETATION À LA HATFIELD–MILGROM

We revisit the algorithm using the characterization $\pi(U - \theta^+) + \theta^- = \bar{\pi}$ on each side.

Period 1, proposal. All matches available, $\mu^{A,0} = M_{xy} := \min(n_x, m_y)$. Since $\nabla G(\alpha) \leq M$, no constraint binds:

$$\tau^{\alpha,1} = 0, \quad \nabla G(\alpha) + R^{\alpha,1} = M.$$

Period 1, disposal. Workers respond:

$$\min\{\tau^{\gamma,1}, \rho^{\gamma,1}\} = 0, \quad \nabla H(\gamma - \tau^{\gamma,1}) + \rho^{\gamma,1} = M - R^{\alpha,1}.$$

Period 1, update. $\mu^{A,1} = M - \rho^{\gamma,1}$.

With cumulative rejected offers $R^{\alpha,t}$ and $R^{\gamma,t}$, initialized $R^{\alpha,0} = M$, $R^{\gamma,0} = 0$:

Period t , proposal: find $(\tau^{\alpha,t}, R^{\alpha,t})$ with $\min\{\tau^{\alpha,t}, R^{\alpha,t}\} = 0$ s.t.

$$\nabla G(\alpha - \tau^{\alpha,t}) = M - R^{\alpha,t} - R^{\gamma,t-1}.$$

Period t , disposal: find $(\tau^{\gamma,t}, R^{\gamma,t})$ with $\min\{\tau^{\gamma,t}, R^{\gamma,t}\} = 0$ s.t.

$$\nabla H(\gamma - \tau^{\gamma,t}) = M - R^{\alpha,t} - R^{\gamma,t}.$$

This collapses to a recursion in the rejection variables alone:

$$\begin{aligned} R^{\alpha,t} &= \rho^{\alpha}(M - R^{\gamma,t-1}), \\ R^{\gamma,t} &= \rho^{\gamma}(M - R^{\alpha,t}). \end{aligned}$$

THEOREM (EQUIVALENCE)

Under continuity/full-support, the quantities $\tilde{\mu}^\alpha(M - R^{\gamma,t})$ and $\tilde{\mu}^\gamma(M - R^{\alpha,t+1})$ from this algorithm coincide respectively with the proposed offers $\mu^{P,t+1}$ and the kept offers $\mu^{K,t+1}$ from the previous algorithm.

Why does this work? The map

$$F(R^\alpha, R^\gamma) = (\rho^\alpha(M - R^\gamma), \rho^\gamma(M - R^\alpha))$$

is **antitone** (because ρ^α, ρ^γ are isotone in their argument by Section 5), so $F^2 = F \circ F$ is **isotone**. Then R^α is a fixed point of the isotone map $R^\alpha \mapsto \rho^\alpha(M - \rho^\gamma(M - R^\alpha))$.

The equilibrium $\mu = \tilde{\mu}^\alpha(M - R^\gamma) = \tilde{\mu}^\gamma(M - R^\alpha)$ satisfies the elegant identity

$$M_{xy} - \mu_{xy} = R_{xy}^\alpha + R_{xy}^\gamma.$$

Section 9

THE SET-VALUED CASE [GHS24]

So far: π is a (single-valued) function $\Rightarrow \mathbf{q}$ is an M-function $\Rightarrow$ unique θ .

But:

- ▶ if ε is degenerate (e.g. deterministic preferences), choice may be *set-valued*;
- ▶ if G^* is not strictly convex, the demand correspondence has flat regions;
- ▶ ties at indifference are generic in finite populations.

Question. Can we generalize MCS results to the set-valued case?

Answer (GHS24). Yes — but the right notion of order is *not* Veinott's strong set order. We need a **dual** object, the *Q-order*, and its associated structure: **matrons**.

This is a theory of MCS for *minimizers of supermodular* (rather than maximizers of supermodular) functions, dual to Topkis.

For functions $\phi, \phi' : \mathbb{R}^Z \rightarrow \mathbb{R}$:

$$\phi \leq_P \phi' \iff \phi(p \wedge p') + \phi'(p \vee p') \leq \phi(p) + \phi'(p') \quad \forall p, p'.$$

Submodular: $\phi \leq_P \phi$.

For sets $B, B' \subseteq \mathbb{R}^Z$:

$$B \leq_P B' \iff \iota_B \leq_P \iota_{B'} \iff \forall p \in B, p' \in B' : p \wedge p' \in B, p \vee p' \in B'.$$

Sublattice: $B \leq_P B$ (Veinott's strong set order).

TOPKIS (1998)

$$\phi \leq_P \phi' \implies \arg \min \phi \leq_P \arg \min \phi'.$$

This handles *maximizers of supermodular* functions; we need the dual.

The natural duality is to ask: what order on ψ 's induces the P-order on conjugates ψ^* 's?

DEFINITION (Q-ORDER)

For convex closed proper l.s.c. ψ, ψ' on $\mathbb{R}^Z$,

$$\psi \leq_Q \psi' \iff \psi'^* \leq_P \psi^*.$$

THEOREM (DIRECT CHARACTERIZATION)

$\psi \leq_Q \psi'$ iff for every $q \in \text{dom } \psi$, $q' \in \text{dom } \psi'$, and every $\delta_1 \in [0, (q - q')^+]$, there exists $\delta_2 \in [0, (q - q')^-]$ with

$$\psi(q - \delta_1 + \delta_2) + \psi'(q' + \delta_1 - \delta_2) \leq \psi(q) + \psi'(q').$$

Exchangeable function: $\psi \leq_Q \psi$.

Reading. A bilateral exchange δ_1 from q to q' on coordinates where $q > q'$ can always be (partly) compensated by a return exchange δ_2 on coordinates where $q < q'$, weakly improving the joint value.

For convex compact $X, Y \subseteq \mathbb{R}^Z$:

$$X \leq_Q Y \iff \iota_X \leq_Q \iota_Y.$$

DEFINITION (MATRON)

X is a *matron* if $X \leq_Q X$, i.e. for any $x, y \in X$ and any $\delta_1 \in [0, (x - y)^+]$, there exists $\delta_2 \in [0, (x - y)^-]$ with

$$x - (\delta_1 - \delta_2) \in X \quad \text{and} \quad y + (\delta_1 - \delta_2) \in X.$$

Comparison.

- ▶ Murota's $M^\natural$ -sets impose δ_1, δ_2 proportional to canonical basis vectors and $\delta_1^\top \mathbf{1} = \delta_2^\top \mathbf{1}$ (combinatorial).
- ▶ Chen–Li's S -convex sets relax the equal-mass condition but still require canonical-basis directions.
- ▶ **Matrons** drop both restrictions: continuous, convex generalization.

THEOREM [GHS24, THM 4]

For c l.s.c. convex proper closed on $\mathbb{R}^Z$, the following are equivalent:

- (1) c is exchangeable, i.e. $c \leq_Q c$;
- (2) c^* is submodular, i.e. $c^* \leq_P c^*$;
- (3) for all $p \leq p'$, $\Pi_{\{p=p'\}}(\partial c^*(p)) \geq_Q \Pi_{\{p=p'\}}(\partial c^*(p'))$;
- (4) defining the rejection correspondence

$$r(p, \bar{q}) = \bar{q} - \arg \max_{q \leq \bar{q}} \{p^\top q - c(q)\},$$

for all $(p, \bar{q}) \leq (p', \bar{q}')$, $\Pi_{\{p=p'\}}(r(p, \bar{q})) \leq_Q \Pi_{\{p=p'\}}(r(p', \bar{q}'))$.

In words. Under exchangeability of c (= submodularity of its conjugate), the rejection correspondence is monotone in $(p, \bar{q})$ *in the set-valued sense given by the Q-order* — on the coordinates where prices coincide.

Quadratic illustration. If $c(q) = q^\top A q / 2$, $A \succ 0$:

c^* submodular $\iff A^{-1}$ is a Stieltjes matrix.

The rejection r and shadow price τ are determined by an LCP:

$$r = S\tau + \rho, \quad r \perp \tau, \quad r, \tau \geq 0.$$

THEOREM (DUAL TOPKIS)

If $\psi + \iota_Q \leq_Q \psi' + \iota_{Q'}$, then

$$\arg \min_{q \in Q} \psi(q) \leq_Q \arg \min_{q' \in Q'} \psi'(q').$$

This is the structural twin of Topkis's theorem, on the Q-side.

Variational preferences. A welfare function $G : \mathbb{R}^{X \times Y} \rightarrow \overline{\mathbb{R}}$ is convex, l.s.c., proper, with $\text{dom } G^*$ compact and $0 = \min(\text{dom } G^*)$. Then

$$G(U) = \max_{\mu \geq 0} \left\{ \sum \mu_{xy} U_{xy} - G^*(\mu) \right\}.$$

DEFINITION [GHS24, DEF. 7]

A *generalized equilibrium matching* between welfare functions G, H with initial preferences α, γ is a triple (μ, U, V) with

- (I) $\max\{U - \alpha, V - \gamma\} = 0$ (money burning)
- (II) $\mu \in \partial G(U) \cap \partial H(V)$ (market clearing)

Condition (i) generalizes the NTU condition $\min\{\tau^\alpha, \tau^\gamma\} = 0$: utility may be *burned* on either side, but not both at once. Connects to [GHJ] on aggregate matching with money burning.

ALGORITHM 1 [GHS24]

At step 0: $\mu_{xy}^{A,0} = \min(n_x^G, n_y^H)$, $\mu^{T,-1} = 0$. Iterate:

Proposal. $\mu^{P,k} \in \arg \max_{\mu^{T,k-1} \leq \mu \leq \mu^{A,k}} \mu^\top \alpha - G^*(\mu)$.

Disposal. $\mu^{T,k} \in \arg \max_{\mu \leq \mu^{P,k}} \mu^\top \gamma - H^*(\mu)$.

Update. $\mu^{A,k+1} = \mu^{A,k} - (\mu^{P,k} - \mu^{T,k})$.

The MCS Theorem of GHS24 (set-valued, Q-order) is exactly what justifies that $\bar{q} \mapsto \bar{q} - M^G(\alpha, \bar{q})$ is isotone in the Q-set order, which is the monotonicity engine of the proof.

THEOREM [GHS24, THM 5]

If G and H are submodular welfare functions, then Algorithm 1 is well-defined and, up to a subsequence, $(\mu^{P,k}, \tau^{P,k}, \tau^{T,k})$ converges. If (μ, τ^P, τ^T) is a limit, and we set

$$\tau^\alpha := \tau^P \mathbf{1}_{\mu < n^G}, \quad \tau^\gamma := \tau^T \mathbf{1}_{\mu < n^H},$$

then $(\mu, \alpha - \tau^\alpha, \gamma - \tau^\gamma)$ is a generalized equilibrium matching with initial preferences α, γ .

Comparison with Alkan–Gale (2003). Their algorithm uses choice functions with *persistence*, a property strictly stronger than Q-isotonicity. Their algorithm converges faster, but ours converges under weaker primitives — exactly the substitutability captured by exchangeability.

Section 10

CONCLUSION

- ▶ **One mathematical engine, three layers of generality:**
 - (I) welfare maximization (Lagrange multipliers, primal/dual)
 - (II) general choice functions (shadow prices, M-functions)
 - (III) set-valued correspondences (Q-order, matrons, dual Topkis)
- ▶ **Substitutability** is the key economic primitive throughout: it appears as Z-property, as M-function, as exchangeability, as conjugate-submodularity — equivalent under suitable regularity.
- ▶ **Existence/uniqueness, comparative statics, and algorithmic convergence** all follow from the same monotonicity story.
- ▶ Concrete payoffs:
 - ▶ closed-form NTU logit (min) versus TU Choo–Siow ($\sqrt{\cdot}$);
 - ▶ aggregate deferred acceptance with continuous quantities;
 - ▶ Hatfield–Milgrom-style monotone reformulation;
 - ▶ generalized equilibria with *money burning*.

Thank you!

Comments and questions welcome.